Software Requirements Specification for Spectropica: games to support the development of individuals with autism

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Revision History

Date	Version	Notes
19/10/2026	1.0	Initial SRS Document

1 Introduction

1.1 Purpose

The purpose of Spectropica is providing a range of games for caregivers, parents or educators to evaluate and develop the skills of autistic learners. The former may tailor the stimulation, difficulty or contents of the games to the needs of the child.

Spectropica is intended to solve the lack of a engaging digital platform for autistic kids to develop their skills. Additionally, we aim to solve the problem that few tools exist for the comprehensive evaluation and progress tracking of autistic learners.

Moreover, physical and digital games designed for autistic kids tend to be rigid and difficult to customize for each student. We aim to solve this issue by allowing caregivers, educators and parents the ability to tailor the games in a variety of ways.

1.2 Scope

The scope of our project is the development of a digital game suite for the evaluation and improvement of the skills of autistic learners under the jurisdiction of caregivers, parents and educators. The intended users are therefore autistic learners with varying skills in varying domains, and Care Team members who utilize features designed to tailor and support the learning path of the child.

The major functionality revolves around the core skill games, and the customization surrounding them. For example, the learning path of the child may involve a vocabulary-based game. The Care Team may customize the game through saving templates to: include words relevant to the child, adjust the colors and stimulative aspects of the game depending on child preferences, change the difficulty based on the child's progress, or decide the reward threshold for cosmetics.

Expected benefits for a player perspective include the boosted engagement of autistic learners in a learning context, the enhancement of their progress in a diversity of skills. From a Care Team perspective, the expected benefits involve the identification of patterns or insights into the child's progress or skill profile through visualized data and numerical metrics that may have been difficult to track beforehand. Additionally, a digitalized platform is ideal for a Care Team member to quickly adapt games to different learners without creating or assembling games from scratch.

To further clarify the scope, Spectropica is intended to be used as a tool for Care Team members rather than a replacement for professional expertise. Within the timeline of Capstone, the software will not retain functionality for Care Team members to create their own brand new game types, design of their own tailored maps for games, or designing cosmetic rewards from scratch. The software will also not include a competitive component for players to compete against each other's scores.

1.3 Stakeholders

- Autistic Learners are a primary stakeholder and an end user group. They play the games and their skill growth is the main objective of the product. Accessibility, engagement and comfort are the priorities for this stakeholder group.

- Customers are secondary stakeholders whoencompass anybody that makes the decision to acquire the software for autistic learners. This includes learning centers adopting the platform for students, parents or caregivers that purchase the software for their child.

- Care Team Members are another end user group that are primary stakeholders. They are the ones familiar with the autistic learner. Consequently, they use both administrative privileges to customize games and the dashboard to read or analyze student-specific data. They are the main decision makers for the autistic learner's learning path and progress, and therefore consulted by the development team for the latter to prioritize the most important tailoring features.

- Domain Experts are the final stakeholder group, falling into the secondary category. They are consulted by the development team to ensure the designed software is appropriate for autistic learners; aiding their progress and steering clear of any unintended harm.

1.4 Definitions and Acronyms

Term	Definition
Learner	Primary player of the games that the care team is aiming to develop the skills and track the progress of.
CTM	Care Team Member. A caregiver, parent, educator or any other figure that is responsible for the learning path of a learner.
Primary CTM	Each Care Team has only one primary CTM for a Learner; the primary CTM has all permissions and is able to manage the permissions of other CTMs.
Template	A pre-defined copy of an existing game that CTMs can create to tailor games for individual students, by modifying the content or stimulation level for example.
Dashboard	A CTM-specific interface designed to visualize data, read key metrics and track the progress of learners.

Table 1: Definitions and Acronyms

1.5 Characteristics of Intended Reader

The intended readers of this SRS are mainly the development team and potentially CTMs and domain experts. The development team will refer back to this SRS throughout the development process to ensure their features align with the outlined requirements and user groups. The CTMs and domain experts may read this SRS as an introduction to understanding Spectropica to be later consulted on domain issues. As such, the intended readers should ideally have a basic understanding of the capabilities and limitations of software, specifically game engines like Godot in the context of the project timeline. In addition, a basic understanding is required of the needs of autistic individuals in a learning environment. This means understanding the most common challenges faced by autistic learners when they engage in learning, what some of the most common preferences are, and what are the boundaries to avoid that may cause harm even unintentionally.

2 System Overview

This section provides an overview of Spectropica, including the problem it addresses, its intended purpose, the environment in which it operates, its users, and the constraints affecting its design. The overview establishes the context for the detailed requirements presented in the following sections.

2.1 Problem Statement

Autistic Learners have different strengths, communication styles, sensory preferences, and support needs (National Institute of Mental Health, 2025). These differences mean that learning activities suitable for one Learner may not be appropriate for another. Caregivers and educators therefore need ways to provide activities that accommodate individual needs rather than relying on a single difficulty level, presentation style, or learning approach.

Providing personalized activities also introduces challenges in managing each Learner's experience. When multiple individuals are involved in supporting a Learner, they need a consistent way to manage learning activities, share relevant information, and coordinate responsibilities. Without a centralized system, maintaining individual configurations and understanding which activities are appropriate for each Learner can become difficult.

Another challenge is interpreting performance over time. A Learner's results may vary depending on the Game, its difficulty, and the conditions under which it was completed. Without organized records of previous Game Sessions, Care Team Members (CTMs) may struggle to identify patterns or make informed decisions about future activities.

Spectropica addresses these challenges through a unified suite of customizable learning Games supported by Learner profiles, CTM-managed configurations, and performance tracking. The system allows CTMs to manage individual Learners, collaborate through care teams, customize Games using Templates, and review recorded performance information. Learners can participate in authorized Games, receive feedback and rewards, and control their participation during gameplay.

Spectropica is intended to provide individualized practice opportunities and support CTMs in understanding performance within the application. It is not intended to provide clinical diagnoses or establish that changes in Game performance represent broader developmental improvements.

2.2 Product Vision

Spectropica aims to provide autistic Learners with an engaging and personalized environment for practising selected skills while giving CTMs a centralized way to manage learning activities and review progress.

The system has five primary objectives:

1. **Support Individualized Learning:** Allow CTMs to manage Learner-specific Game authorizations, difficulty levels, personalized settings, and Templates so that activities can reflect individual needs and preferences.

2. **Enable Coordinated Care Team Management:** Provide a structured way for multiple CTMs to support the same Learner while maintaining appropriate permissions and responsibilities. A Primary CTM manages access to the Learner's information and configuration.

3. **Provide Accessible and Engaging Gameplay:** Offer Games with understandable instructions, opportunities for repeated practice, and rewards intended to encourage engagement. Learners retain the ability to pause, restart, or end activities.

4. **Support Meaningful Performance Tracking:** Record Game Session results and provide authorized CTMs with access to individual Game histories, performance representations, and data exports.

5. **Maintain Consistent Learner Experiences:** Allow Learners to access their profiles through paired devices while preserving their assigned Games, configurations, and associated information.

The initial learning areas are routine sequencing, functional communication, and short-term memory. These areas establish the intended focus of the initial Games, while Templates allow CTMs to customize existing Games rather than create entirely new Game types.

These objectives define Spectropica's intended benefits without prescribing the implementation of individual features. Together, they support a learning environment in which CTMs can personalize activities, coordinate responsibilities, and review information while Learners participate in Games suited to their needs.

2.3 System Context

Spectropica is intended for use in environments where autistic Learners participate in learning activities with varying levels of support from caregivers or educators. These environments may include homes and learning centres.

The system interacts with two primary user groups: Learners and CTMs. Learners access the Game Suite to participate in authorized Games, while CTMs use the Dashboard to manage Learner profiles, Game configurations, Templates, care team permissions, and performance information.

Users interact with Spectropica through supported devices, with tablet applications for iPadOS and Android forming the primary intended operating environments. Web access is also planned. The system is being developed using Godot, whose documentation describes exports for Android, iOS, and web browsers (Godot Engine contributors, n.d.). Support for a particular platform depends on the team's ability to implement and test it.

Devices provide the input, display, and connectivity capabilities needed to access Spectropica. The system manages the association between Learners and their paired devices.

The system boundary includes Spectropica's application functionality running on supported devices and its server-side functionality responsible for managing accounts, permissions, configurations, Games, and records. The devices, operating systems, browsers, and network connectivity used to access Spectropica form part of its operating environment rather than separate application components.

The system manages information associated with CTM accounts, Learner profiles, care team memberships, Games, Game Sessions, Templates, and rewards. These entities are internal to Spectropica and are described in Section 7.

Figure 1 illustrates Spectropica's system boundary and the information exchanged with its external entities: Learners and CTMs. The arrows represent the direction of information flow rather than the order in which operations occur.

Learner Responsibilities Learners provide Game selections, responses to activities, and participation choices. They may also interact with rewards and cosmetic items made available through the Game Suite. Learners are responsible for making selections during gameplay but are not expected to manage their own configurations or permissions.

CTM Responsibilities CTMs provide information needed to create and manage Learner profiles, configure Games, customize Templates, and interpret performance results. Authorized CTMs may manage Learner information according to their assigned permissions. The Primary CTM is responsible for managing the membership and permissions of other CTMs associated with a Learner.

CTMs are responsible for choosing configurations and content appropriate to the Learners they support. Spectropica does not independently determine the suitability of a learning activity for an individual Learner.

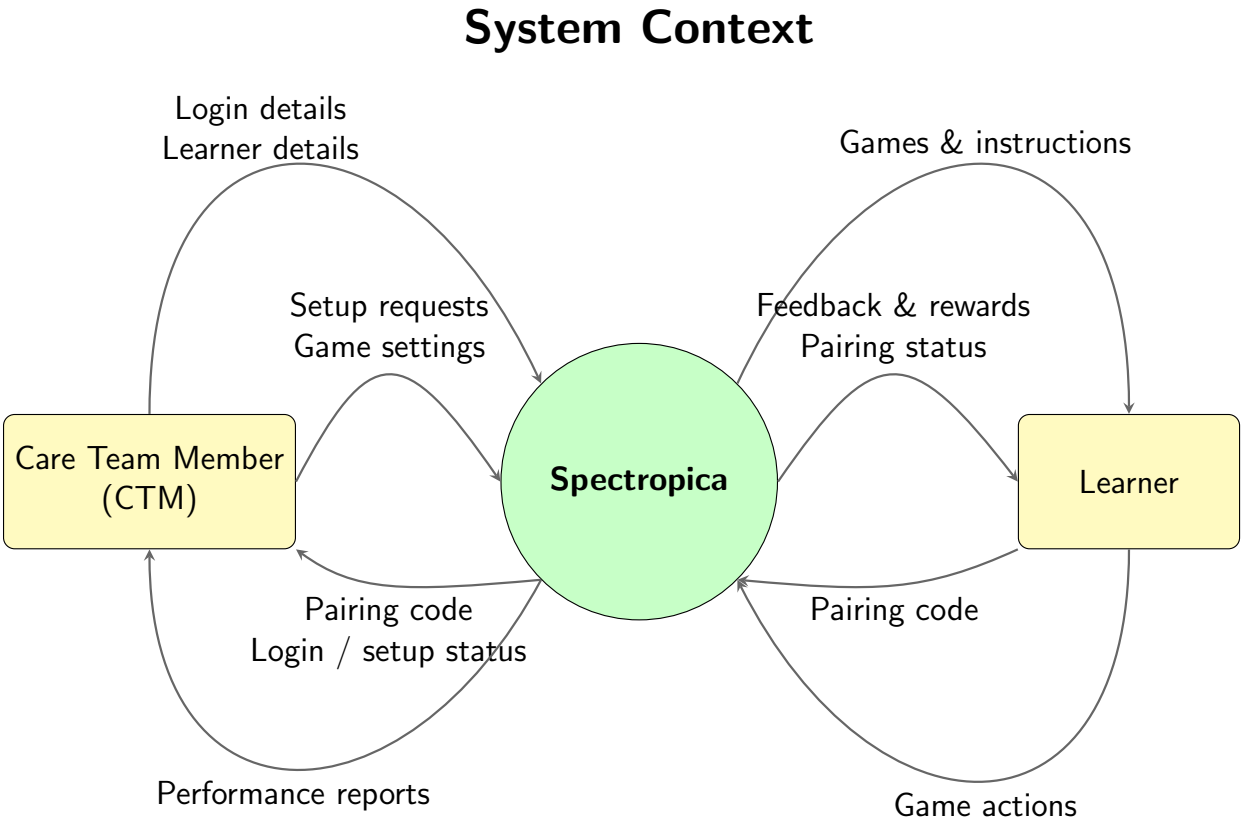


Figure 1: System Context Diagram for Spectropica

Supported Device and Network Responsibilities The device provides the input and display capabilities through which users interact with Spectropica. Network connectivity supports communication with the system and synchronization of Learner information.

External Email-Delivery Service Responsibilities An external email-delivery service supports the account-verification functionality specified in FR-CTM-3. The service is responsible for delivering verification messages requested by Spectropica, while Spectropica remains responsible for managing account verification. No specific email provider is prescribed.

Spectropica Responsibilities Spectropica is responsible for receiving and validating user inputs, managing CTM accounts and Learner profiles, maintaining device pairings, and enforcing access permissions.

During gameplay, the system is responsible for presenting authorized Games with the appropriate Learner Game Configuration or assigned Template. It also provides feedback, manages rewards and cosmetic items, and records Game Session information.

For CTMs, Spectropica provides access to permitted management functions and representations of Learner performance information. It maintains the relationships between Learners, CTMs, Games, configurations, and session records.

Spectropica is also responsible for temporarily retaining unsynchronized Game Session results when network connectivity is unavailable, as specified in FR-DATA-8.

The system requests delivery of account-verification messages through the external email-delivery service as part of the CTM registration process.

Spectropica is an educational support system rather than a clinical or safety-critical application. Its intended use involves providing opportunities for practice and presenting information about performance within the Games.

Although it is not a clinical system, Spectropica manages information about potentially vulnerable users. Appropriate access controls and handling of Learner information are therefore important to its operation.

The external email-delivery service supports account verification through communication beyond Spectropica's application boundary. This does not prescribe a particular email provider or implementation technology.

2.4 User Characteristics

Spectropica has two primary user groups: Learners and Care Team Members (CTMs). Each group has different responsibilities, technical abilities, and interaction needs.

Learners Learners are autistic individuals who use Spectropica to participate in Games and practise selected skills. The initial target population identified by the project is children aged eight to ten, although suitability depends on individual abilities and support needs rather than age alone.

Autistic individuals have varied communication characteristics and may be more or less sensitive to sensory input (National Institute of Mental Health, 2025). For design purposes, the project also assumes that Learners may vary in attention, reading skills, motor coordination, and experience with digital devices and Games.

The intended Learner should be able to make intentional selections using a supported interaction method, independently or with assistance. Independent reading, spoken communication, and advanced technical knowledge are not assumed.

Learners are expected to interact primarily with the Game Suite. Their tasks include selecting authorized Games, responding to instructions, controlling gameplay, and interacting with available rewards or cosmetic items. They are not expected to understand account management, permission structures, or performance analytics.

Care Team Members (CTMs) CTMs include parents, guardians, caregivers, educators, and other individuals involved in supporting a Learner's activities.

CTMs are expected to have basic digital literacy, including the ability to navigate an application, create or access an account, enter information, and manage settings. Programming experience or knowledge of software development is not required.

CTMs are expected to have sufficient familiarity with the Learner's needs and preferences to select appropriate Games, adjust configurations, and interpret recorded performance information.

CTMs may have different levels of authority within a Learner's care team. The Primary CTM manages other members' permissions, while other CTMs can access functions according to the permissions assigned to them.

Domain Experts Domain Experts, such as special-education professionals and autism researchers, are stakeholders who may provide guidance on the appropriateness of learning activities and the interpretation of results. They are not defined as a separate application user role in the current functional requirements.

2.5 System Constraints

The following constraints limit the operating environment and design choices for Spectropica. They are distinguished from functional and nonfunctional requirements, which describe the system's behavior and quality attributes in later sections.

SC1 — Supported Operating Platforms Spectropica is intended to support tablet applications on iPadOS and Android, as well as access through a web browser. Support for additional desktop operating systems, including Windows and macOS, may be considered depending on implementation and testing feasibility. Linux support is not a priority for the initial release.

SC2 — Supported Input Hardware The initial system is intended to support touch input on compatible tablets. Additional input methods, including keyboard and pointer

interaction, may be supported where appropriate for the selected deployment platforms.

SC3 — Networked Account and Data Management
Network connectivity is required for operations involving shared account information, care team membership, and synchronization of Learner data. FR-DATA-8 specifies temporary retention of unsynchronized Game Session results but does not establish full offline operation.

3 Stakeholder Requirements

Autistic Learners

SR-LRN-1: Games are not overstimulating or understimulating for Learners with sensory hypersensitivity or hyposensitivity, respectively (Forbrain, 2026)

SR-LRN-2: Games are unambiguous with clear rules and instructions (FRATNOW, 2025)

SR-LRN-3: Games are a safe space for Learners, allowing them to take their time and retry if needed (FRATNOW, 2025)

SR-LRN-4: Games are fun and engaging for Learners and give them a sense of accomplishment

Customers

SR-CUST-1: System is easy to install, implement, and manage for varying levels of technical literacy

SR-CUST-2: System is highly reliable from a technological point of view

SR-CUST-3: System must be safe for autistic children to use and avoid elements that can trigger emotional or physical distress in Learners

Care Team Members

SR-CTM-1: Game content is educational and aids Learners with their skill development

SR-CTM-2: Gameplay settings such as noise can be configured for a Learner's specific needs

SR-CTM-3: Aspects of game content (such as a vocabulary list in a reading game) can be tailored towards a specific Learner

SR-CTM-4: Learners' progression in games can be viewed and tracked over time

SR-CTM-5: Multiple CTMs are able to collaborate in managing a Learner's profile

SR-CTM-6: The primary CTM can manage other CTMs' permissions on a Learner's profile and control what can be accessed and changed

Domain Experts

SR-DOM-1: Educational aspect of game content is in line with intended age group's school curriculum

SR-DOM-2: Games interest Learners and retain their attention

SR-DOM-3: Games consider and support idiosyncracity of autistic individuals' sensory processing

SR-DOM-4: Games are not static; they allow for progression of difficulty

Note that there is some natural overlap between stakeholders' requirements. Requirements were categorized under what the development team considers the most relevant stakeholder to them.

4 User Personas

4.1 Persona 1

- Name: Layla
- Description: Layla is a 9-year-old autistic learner who enjoys speaking her mind. Layla is a strong academic student that excels in mathematics and reading. Layla also struggles with social interactions and has difficulty understanding social cues which hamper her ability to make friends and socialize with peers or the public. Layla can be easily overstimulated by loud noises, bright lights or fast movements.
- Goals: Layla wants to make friends easier and conversate without being scared or shy. She also wants to express herself and understand other people's expressions.
- Frustrations: Layla finds friends hard to make. She feels misunderstood when she expresses herself. Games can often contain: loud noises, bright lights and cluttered screens that can be overwhelming.
- Technical Ability: Layla is a tech-savvy child that enjoys playing video games on her laptop. She is comfortable with basic computer functions such as dragging and dropping, clicking and typing.

4.2 Persona 2

- Name: Adam
- Description: Adam is a 6-year-old autistic learner who enjoys being social and chatty with his friends. Adam struggles with executive functioning and has difficulty focusing on academic tasks for extended periods of time, especially in mathematics and reading. Adam can be quickly understimulated by repetitive tasks and become bored.

- Goals: Learn math without being bored. Find reading easy. Focus on academic activities for longer.
- Frustrations: Adam thinks math is boring and reading is hard, and that both take too long without any stimulation. Educational games lack rewards or any incentive to complete.
- Technical Ability: Adam does not have access to a laptop, but is comfortable on touch screen devices such as his tablet. He does not like flat-looking designs with nothing particularly engaging happening.

4.3 Persona 3

- Name: Noah
- Description: Noah is a 65-year-old licensed educator at an Autism Learning Center. Noah teaches classrooms of 10 autistic learners at a time, often in the 4-10 year old range. Noah also teaches autistic learners 1-on-1 across a range of skills including social skills, communication, mathematics and reading. Noah uses physical materials either hand-crafted or externally purchased to facilitate learning activities with his students.
- Goals: Noah wants to track the progress of his diverse set of students individually across different skills. Noah wants a more malleable method of tailoring his activities for each student's preferences.
- Frustrations: Noah finds that having several students can strain the time available to analyze each one's specific progress, whether they have made any and to what extent. Noah likes the physical materials employed in his activities, but they are not malleable enough to be personalized for each student's experience.
- Technical Ability: Noah's formative years in the world of teaching autistic learners preceded widespread technology. Therefore, Noah is able to use tablets and computers at a basic level, but it may be confusing with more complicated functionality or presentation.

4.4 Persona 4

- Name: Sara
- Description: Sara is a 32-year-old mother of a 6-year-old autistic child. Sara's job in the engineering industry enables her to quickly adopt new technologies and use them proficiently. She is experienced at reading and analyzing data. However, Sara's knowledge in the field of autism is limited.

- Goals: Sara wants to aid her child's learning path. Sara knows her child's preferences and habits well, and wants to create a curated experience for her child based on this knowledge.
- Frustrations: Sara is unsure which activities are suitable for the skill development of autistic children. When she does find a pre-packaged activity, her child tends to not respond well because the experience is not personalized.
- Technical Ability: Sara is proficient at hugely different devices and technologies. She is comfortable with laptops, tablets and reading visualized or numerical data.

5 Use Cases

5.1 UC1: Play a game

Primary Actor

Autistic Learner

Preconditions

- A CTM has authorized that the Learner can play this game.
- A CTM has either approved default settings or configured personalized settings for this game for the learner.

Main Success Scenario

1. System's game suite displays games Learner can play.
2. Learner selects a specific game from their game suite.
3. System loads the selected game with the correct settings.
4. Learner chooses to start the game when they are ready.
5. System continues presenting tasks completed by the Learner.
6. After the final task is completed, System loads the final completion screen.
7. System saves the Learner's performance data from the session into the correct database(s).
8. Learner claims their in-game reward.
9. Learner exits the game.

Alternative Flows

- 1.1: System displays that the Learner cannot play any games at the moment.

- 2.1: The system fails to load the selected game with the correct setting.
  2.2: CTM or Learner aborts the game.
- 3.1: Learner chooses to return to game suite screen.
- 4.1: Learner finds instruction unclear.
  4.2: Learner wants a Guide for this instruction.
  4.3: System loads Guide.
  4.4: Learner closes Guide.
- 6.1: System does not find a stable connection to the database(s).
  6.2: System queues the data for future syncing.
- 8.1: Learner or CTM start the game again.

5.2 UC2: Adding a Learner

Primary Actor

CTM

Preconditions

- The CTM is logged into their account.

Main Success Scenario

1. CTM chooses to Add Learner.
2. System loads the configuration page for a new learner's information.
3. CTM inputs Learner information.
4. CTM chooses which games this Learner can play, with configurable per-game settings.
5. System displays a special Learner code for pairing purposes.
6. The special code is entered correctly by an actor on the Learner's device(s).
7. System saves the Learner's information to the appropriate tables in the database.
8. System visually confirms successful pairing on the CTM account (Learner added) and the Learner's device(s).

Alternative Flows

- 1.1: CTM chooses to transfer Learner profile to their account from another CTM.
  1.2: Other CTM account confirms transfer of profile ownership.

- 6.1: Actor's code input was incorrect.

  6.2: System allows for a limited amount of retries before blocking more tries for a period of time.

  6.3: CTM chooses to generate new code for this Learner.

Postconditions

- Learner profile has been created in relationship with the CTM's account.
- Learner can see the correct Learner screen on their paired device(s).
- The CTM that created the Learner profile is the primary CTM for this Learner.

5.3 UC3: View Learner data

Primary Actor

CTM

Preconditions

- The CTM is logged into their account.
- The CTM is authorized to view this Learner's profile.

Main Success Scenario

1. CTM chooses to view a specific Learner's data.
2. System loads Learner's data and displays the main page.
3. CTM views the Learner's most recent game sessions.
4. CTM views an overview of the Learner's average performance on each game.
5. CTM chooses a specific game to view more specific details.
6. System loads the Learner's statistics and session history for this game.
7. CTM chooses a view (eg. table, chart) for the Learner's statistics in this game such as overall score and where the Learner's strongest and weakest performance during sessions typically are.
8. CTM views a data representation of the Learner's overall score in this game across their timeline since their first session.
9. CTM exports the Learner's data into a different format.
10. CTM exits the Learner data page.

Alternative Flows

- 3.1: CTM chooses to view a specified portion of the Learner's overall game sessions (eg. October 6 - October 15).
- 8.1: CTM sets a specific timeline to view the Learner's score such (eg. October 6 - October 15).

Postconditions

- CTM has viewed the requested data successfully.

5.4 UC4: Personalize Learner Settings

Primary Actor CTM

Preconditions

- The CTM is logged into their account.
- The CTM is authorized to change this Learner's settings.

Main Success Scenario

1. CTM chooses the Learner's profile.
2. CTM chooses a specific game to personalize Learner settings for.
3. CTM picks specific difficulties of this game to be available for this Learner.
4. CTM picks a specific stimulation level.
5. CTM adjusts reward for completing the game successfully.
6. CTM chooses prompt level such as full help, partial hints or fully independent.
7. CTM chooses volume levels for music, sound effects and voice.
8. CTM chooses intensity of celebrations when the Learner is successful.
9. CTM saves new settings for this Learner on this game.

Alternative Flows

- 3.1: CTM allows all difficulties to be available for this Learner on this game.
- 4.1: CTM does not change the default stimulation level.
- 5.1: CTM does not adjust the baseline rewards.

- 6.1: CTM does not change the default prompt level.
- 7.1: CTM does not change the default volume levels.
- 8.1: CTM does not change the default intensity.

Postconditions

- System has successfully saved the personalized Learner's settings for this game.
- Learner can only play the game with personalized settings.

5.5 UC5: Adding Game Template

Primary Actor CTM

Preconditions

- The CTM is logged into their account.

Main Success Scenario

1. CTM chooses a specific game to create a template out of.
2. System loads the template screen.
3. CTM chooses to add, change or remove contents within the game. Example: changing 'Cereal' to a specific brand, removing candy or adding chocolate in a grocery-based game.
4. CTM chooses to adjust the colour palette, volume levels, intesntiy for this game.
5. CTM chooses a difficulty for this template.
6. CTM chooses the Learner(s) this template should be assigned to.
7. CTM chooses to export the template to a file format.
8. System replaces the default version of the game with the template version for the assigned Learner(s).
9. System displays a success message for the CTM.

Alternative Flows

- 1.1: CTM imports template from another CTM.
- 3.1: CTM chooses to not change the content.
- 4.1: CTM chooses to go with the default settings.

- 6.1: System says this Learner already has a template for this game and asks whether to replace that assignment.

  6.2: CTM saves the template under a unique name without assigning it yet.

  6.2: CTM saves the template under a specific folder in their templates screen.

- 9.1: System displays a failure message for the CTM with reasoning.

Postconditions

- CTM has successfully created, assigned and saved the template.
- System has saved which Learners play the created template.
- Learner can play the assigned template version of the game.

5.6 UC6: Equipping Cosmetic Item

Primary Actor Learner

Preconditions

- The Learner is logged into their profile.

Main Success Scenario

1. Learner chooses to open the Items screen.
2. System loads the item page with any filters applied by the CTM i.e to hide certain items.
3. Learner chooses a cosmetic item to buy.
4. System grants the cosmetic item to the Learner.
5. The Learner chooses to equip their cosmetic item.
6. System updates the Learner's view to show the equipped cosmetic item (eg. shirt, pet).
7. Learner exits the Item screen.

Alternative Flows

- 3.1: Learner equips a cosmetic item they already have.
- 4.1: System finds that the cosmetic item cannot be unlocked.

  4.2: System displays reasoning for failure to unlock.

  4.3: System explains when it is possible to unlock this item.

  4.4: System shows possible alternatives to the wanted item.

- 5.1: Learner chooses to not equip their item.
- 7.1: Learner chooses to browse further items.

Postconditions

- Learner screen displays their updated view with the equipped item(s).
- System has saved the newly equipped items to the current view.
- System has saved the newly unlocked items to what items the Learner owns.
- If applicable, System has deducted currency from the Learner's balance.

5.7 UC7: Create CTM account

Primary Actor CTM

Preconditions

- CTM is on the registration page.
- The CTM's device is connected to the Internet.

Main Success Scenario

1. CTM enters their name, email and password.
2. CTM chooses from a collection of default avatars.
3. CTM agrees to terms & conditions.
4. System saves entered information and agreement into the appropriate data tables.
5. System sends the CTM a verification email.
6. CTM verifies their email.
7. System creates CTM account.
8. System sends CTM an email confirming their account creation.
9. CTM uses previously entered information to log into their account.

Alternative Flows

- 4.1: System prompts CTM to agree to terms & conditions if they did not explicitly agree before submitting.

  4.2: System prompts CTM to log in after finding the email is associated with an existing account.

  4.3: System prompts CTM to choose a stronger password after finding the password too weak.

  4.3: System prompts CTM to check their re-typed password after finding the passwords do not match.

- 5.1: CTM asks to re-send verification email after expiry.

  5.2: System re-sends verification email.

Postconditions

- System has successfully saved the newly created CTM account into the database.
- CTM is able to successfully log in with the information they previously entered during creation.

5.8 UC8: Delete Learner Profile

Primary Actor CTM

Preconditions

- CTM is authorized to delete this Learner's profile.

Main Success Scenario

1. CTM chooses the intended Learner's profile.
2. CTM chooses to delete Learner's profile.
3. System explains what this entails and offers to export the Learner's data before deletion.
4. CTM confirms deletion of Learner's profile.
5. System unlinks Learner's paired devices.
6. System revokes access to other CTM's that were able to view this Learner's profile.
7. System schedules account's permanent deletion after a short grace period.
8. System sends notification to CTM as a confirmation of deletion.

Alternative Flows

- 4.1: CTM chooses to not delete Learner's profile.
- 6.1: CTM decides to restore Learner's profile during the grace period.
  6.2: System restores Learner data and cancels scheduled deletion.

Postconditions

- System has successfully scheduled the Learner's profile for deletion.
- System does not retain any Learner data after deletion for privacy reasons.
- Learner can no longer enter their deleted profile.

5.9 UC9: Add CTM to Learner

Primary Actor Primary CTM

Preconditions

- The Primary CTM is logged into their account.
- The Learner's profile exists.

Main Success Scenario

1. Primary CTM chooses the Learner's profile.
2. Primary CTM chooses to add a CTM to the Learner's care team.
3. Primary CTM selects the permissions the New CTM will have (eg. view Learner data, personalize settings, assign templates).
4. System generates an invite code with an expiry time.
5. Primary CTM shares the invite code with the new CTM.
6. New CTM enters the invite code on their account.
7. System validates the invite code.
8. System adds the New CTM to the Learner's care team with the selected permissions.
9. System displays the Learner on the New CTM's account.
10. System notifies the Primary CTM that the New CTM has joined.

Alternative Flows

- 3.1: Primary CTM keeps the default permissions.

- 5.1: Primary CTM cancels the invite before it is used.
- 6.1: New CTM does not have an account.
- 7.1: Invite code is incorrect, expired or already used.

  7.2: System allows a limited number of retries before blocking further attempts for a period of time.

  7.3: Primary CTM generates a new invite code.
- 8.1: New CTM is already part of this Learner's care team.

Postconditions

- The new CTM can see the Learner's profile on their account.
- New CTM is limited to the pre-selected permissions by the primary CTM.

5.10 UC10: Manage CTM Permissions

Primary Actor

Primary CTM

Preconditions

- The Primary CTM is logged into their account.
- The Learner's care team has at least one CTM other than the Primary CTM.

Main Success Scenario

1. Primary CTM chooses the Learner's profile.
2. Primary CTM chooses to view the Learner's care team.
3. Primary CTM selects a CTM.
4. System displays the CTM's permissions and actions related to this Learner.
5. Primary CTM grants, edits or revokes permissions for the selected CTM.
6. Primary CTM saves the changes.
7. System applies the new permissions immediately.
8. System notifies the affected CTM that their permissions have changed.

Alternative Flows

- 5.1: Primary CTM chooses to remove the selected CTM from the care team.

  5.2: System asks the Primary CTM to confirm the removal.

  5.3: System revokes the CTM's access to the Learner and notifies them.

- 6.1: Primary CTM chooses to no apply the changes.

Postconditions

- The selected CTM's permissions are changed.
- There is exactly one Primary CTM for the Learner.

6 Functional Requirements

textbfCTM Accounts

FR-CTM-1: The system shall check a CTM's account registration to validate entered information.

FR-CTM-2: The system shall display reasons for errors in the CTM registration process.

FR-CTM-3: The system shall create the CTM's account after they verify their email.

FR-CTM-4: The system shall allow a CTM to log in using the information verified in the registration process.

FR-CTM-5: The system shall allow a CTM to delete their account with a scheduled grace period, un-linking all Learners and templates.

FR-CTM-6: The system shall allow a CTM to restore their account during the grace period.

Learner Profiles

FR-LRN-1: The system shall allow a CTM to create a new Learner profile under their account.

FR-LRN-2: The system shall designate the CTM that created the Learner profile as the primary CTM.

FR-LRN-3: The system shall allow a CTM to input the Learner's specific information.

FR-LRN-4: The system shall allow the primary CTM to delete a Learner's profile with a scheduled grace period.

FR-LRN-5: The system shall allow the primary CTM to export a Learner's data prior to deletion.

FR-LRN-6: The system shall delete all Learner-related data after the deletion grace period.

FR-LRN-7: The system shall allow the primary CTM to restore a Learner's profile during the grace period.

FR-LRN-8: The system shall not allow a Learner to access their deleted profile.

Device Pairing

FR-DEV-1: The system shall allow a Learner to pair multiple devices.

FR-DEV-2: The system shall confirm the successful pairing of each device to both parties.

FR-DEV-3: The system shall allow a CTM to generate a new code for the Learner's device pairing.

FR-DEV-4: The system shall allow a specific amount of retries before the pairing method is nullified.

FR-DEV-5: The system shall block further pairing attempts for a period of time after the retry limit is reached.

Care Team & Permissions

FR-TEAM-1: The system shall allow the primary CTM to add other CTM's to the Learner's care team via invite code.

FR-TEAM-2: The system shall allow the primary CTM to cancel an outgoing invite.

FR-TEAM-3: The system shall allow the primary CTM to generate a new invite.

FR-TEAM-4: The system shall allow the primary CTM to edit the permission of other CTMs in the Learner's care team.

FR-TEAM-5: The system shall allow the primary CTM to remove another CTM from the Learner's care team.

FR-TEAM-6: The system shall allow a CTM to transfer ownership of a Learner's profile from another CTM.

FR-TEAM-7: The system shall not allow a CTM to perform actions they do not explicitly have permission to do.

Personalization

FR-PERS-1: The system shall allow a CTM to toggle the authorization of specific games for a Learner.

FR-PERS-2: The system shall allow an authorized CTM to toggle the authorization of specific difficulties of a game for a Learner.

FR-PERS-3: The system shall allow an authorized CTM to choose specific personalized settings of a game for a Learner.

FR-PERS-4: The system shall always load the personalized settings for a game when a Learner plays it, unless a template is assigned.

Templates

FR-TMPL-1: The system shall allow an authorized CTM to create a new template from an existing game in their saved templates.

FR-TMPL-2: The system shall allow an authorized CTM to import a template from another CTM.

FR-TMPL-3: The system shall allow a CTM to add, change or remove default contents within a template.

FR-TMPL-4: The system shall allow a CTM to tailor settings for this template.

FR-TMPL-5: The system shall allow a CTM to assign a difficulty for a template.

FR-TMPL-6: The system shall allow a CTM to assign the template to Learner(s), overriding any previous settings.

FR-TMPL-7: The system shall allow a CTM to un-assign a template to Learner(s), restoring previous settings.

item[FR-TMPL-8:] The system shall allow a CTM to delete a template which un-assigns it from any associated Learner(s).

FR-TMPL-9: The system shall allow a CTM to create folders that can hold more than one template.

Gameplay

FR-PLAY-1: The system shall not allow the Learner to play the games they are not explicitly authorized to play.

FR-PLAY-2: The system shall not start a game until it receives a cue that the Learner is ready.

FR-PLAY-3: The system shall only present the next task once it verifies the Learner has completed the current task.

FR-PLAY-4: The system shall allow the Learner to read a guide regarding the current instruction if permitted by the CTM.

FR-PLAY-5: The system shall allow the Learner to pause an ongoing game.

FR-PLAY-6: The system shall allow the Learner to restart an ongoing game.

FR-PLAY-7: The system shall allow the Learner to abort an ongoing game.

FR-PLAY-8: The system shall allow the CTM to abort a Learner's ongoing game.

FR-PLAY-9: The system shall allow the Learner to play again after a completed game if permitted by the CTM.

FR-PLAY-10: The system shall allow the Learner to return to the game suite screen after a completed game.

Rewards & Cosmetics

FR-RWD-1: The system shall allow a Learner to claim an in-game reward after their completed game.

FR-RWD-2: The system shall not offer duplicate items as rewards.

FR-RWD-3: The system shall allow a Learner to view cosmetic items in the items screen.

FR-RWD-4: The system shall allow a Learner to unlock an item they meet the conditions for.

FR-RWD-5: The system shall allow a Learner to equip an item they already own.

FR-RWD-6: The system shall display a reason for when an item cannot be unlocked while outlining the conditions for unlocking.

FR-RWD-7: The system shall display alternatives for an item that could not be unlocked when buyable alternatives exist.

Learner Data

FR-DATA-1: The system shall allow an authorized CTM to view a Learner's general data.

FR-DATA-2: The system shall display an expandable overview of the Learner's game history.

FR-DATA-3: The system shall allow an authorized CTM to view a Learner's game history between specific dates.

FR-DATA-4: The system shall allow an authorized CTM to view a Learner's game-specific data.

FR-DATA-5: The system shall offer multiple data representations of the Learner's game-specific data.

FR-DATA-6: The system shall allow an authorized CTM to export a Learner's game data.

FR-DATA-7: The system shall save the Learner's performance in the game session into the correct data tables.

FR-DATA-8: The system shall queue the Learner's performance locally for future syncing if it does not have a stable connection.

FR-DATA-9: The system shall not gather any data more than what is required to fulfill the purpose of loading the correct game configurations, cosmetic items and uploading session data.

Safety Requirements

The following were generated from the hazard analysis:

SR-1 The system shall enforce password strength on creation of account.

SR-2. The system shall end a CTM's session after a period of inactivity.

SR-3. The system shall require a CTM to re-enter their password before deleting a Learner profile, deleting their account, transferring ownership of a Learner profile or removing a CTM from a care team.

SR-4. Default stimulation level, volume and celebration intensity shall be set to low values until a CTM changes them.

SR-5. No game audio or animation shall begin before the Learner gives the cue to start.

SR-6. Games shall not contain content that flashes more than three times in any one-second period (following WCAG 2.x Success Criterion 2.3.1).

SR-7. Error messages shown to a Learner shall be calm and short, shall not use alarming sounds or visuals, and shall offer a single action to return to the game suite.

SR-8. Sessions that end through a crash, abort or lost connection shall be recorded as incomplete and shall not be counted in a Learner's performance averages.

SR-9. A loss of connection shall not interrupt a game session in progress or prevent a Learner from playing authorized games with settings already stored on the device.

SR-10. Every game task shall be completable with either touch or mouse input.

SR-11. Pause, restart and exit controls shall be visible and responsive at all times while a game is being played.

SR-12. Cosmetic items hidden by a CTM shall be hidden from the items screen and shall not be purchasable or equippable by the Learner.

SR-13. The system shall confirm a save of settings, templates or equipped items only after the write has been verified; if saving fails, the previous values shall be kept and the CTM or Learner informed.

SR-14. Before a template is assigned, un-assigned or deleted, the system shall show the affected Learners and what will be replaced, require confirmation, and show a preview of the template.

SR-15. Imported files shall be treated as untrusted data and shall never be executed.

SR-16. An export of Learner data shall contain every record for the selected Learner, game and date range, and shall report a record count that matches the dashboard for the same selection.

SR-17. The system shall automatically back up stored data regularly and shall support restoring from a backup.

SR-18. The system shall verify that deletion of a Learner's data has completed and shall remove the data from backups within a fixed retention period (defined in the SRS Appendix of auxiliary constants).

7 Data Requirements

7.1 Major Data Entities

- Learner: The profile of a Learner and their paired devices.
- Device: information pertaining to a Learner's paired devices.
- CTM: The registered CTM account with login credentials and template folders.
- Care Team Membership: Each link between a CTM and a Learner, the permissions associated and whether they're the primary CTM.
- Game: A game object with its information and default settings.
- Game Session: A play-through session of a Learner on a specific game with scores and other relevant statistics.
- Learner Game Configuration: The tailored per-game configuration for a Learner, including personalized settings, assigned templates and authorizations.
- Template: A CTM-made variation of an existing game with customized content and/or settings.
- Cosmetic Item: an item (eg. shirt, pet) that can be unlocked or earned.
- Learner Inventory: a collection of cosmetic items owned by the Learner, when they were owned and whether they're equipped.
- Access Code: a time-limited unique code, whether it's for device pairing or CTM invites, and tracking attempts for lockout.

7.2 Data Dictionary

Attribute	Type	Description
learnerId	UUID	Unique Learner identifier
createdAt	DateTime	When the profile was created

learnerName	String	Learner name displayed in software
deletionTime	DateTime	When deletion is scheduled, or null

Table 2: Learner

Attribute	Type	Description
deviceId	UUID	Unique device identifier
learnerId	UUID	Unique Learner identifier
deviceName	String	Readable name for the device
pairedAt	DateTime	When the device was paired
isActive	Boolean	Whether the pairing is active

Table 3: Device

Attribute	Type	Description
ctmId	UUID	Unique CTM identifier
createdAt	DateTime	When the profile was created
ctmName	string	CTM Name displayed in software
ctmEmail	String	Unique email address
emailVerified	Boolean	Whether email is verified
passwordHash	String	Securely hashed password
termsAcceptedAt	DateTime	Time of terms acceptance
deletionTime	DateTime	When deletion is scheduled, or null

Table 4: CTM

Attribute	Type	Description
careTeamId	UUID	Unique Care Team identifier
learnerId	UUID	Unique Learner identifier
ctmId	UUID	Unique CTM identifier
isPrimary	Boolean	Whether the CTM is the primary CTM for this learner
permissions	Enum	Permission granted by the CTM regarding this learner's profile
joinedAt	DateTime	Time of joining the Learner's care team

Table 5: Care Team Membership

Attribute	Type	Description
gameId	UUID	Unique Game identifier
gameName	String	Game display name
gameDesc	String	Description of game
defaultSettings	JSON	Settings applied by default

Table 6: Game

Attribute	Type	Description
sessionId	UUID	Unique session identifier
learnerId	UUID	Unique Learner identifier
gameId	UUID	Unique game identifier
templateId	UUID	Template used; null if none
settingsUsed	JSON	Settings active during the session
startedAt	DateTime	When the session started
endedAt	DateTime	When the session ended
outcome	Enum	Completed, aborted, restarted
scores	JSON	Individual task scores and overall score

Table 7: Game Session

Attribute	Type	Description
configId	UUID	Unique configuration identifier
learnerId	UUID	The Learner this configuration applies to
gameId	UUID	The Game this configuration applies to
isAuthorized	Boolean	Whether the Learner can play the Game
settings	JSON	Personalized settings and authorizations for this Learner on this game
reward	String	Reward given for completing the Game
templateId	UUID	Template assigned, null if none
updatedBy	UUID	CTM who last changed the configuration
updatedAt	DateTime	When the configuration was last changed

Table 8: Learner Game Configuration

Attribute	Type	Description
templateId	UUID	Unique template identifier

ownerCtmId	UUID	CTM who has this template
gameId	UUID	Game the template is based on
templateName	String	Template display name
folderName	String	Folder the template is organized under; null if unfiled
modifiedContent	JSON	Edited game content
settings	JSON	Template settings
createdAt	DateTime	When the template was created
updatedAt	DateTime	When the template was last changed

Table 9: Template

Attribute	Type	Description
itemId	UUID	Unique item identifier
itemName	String	Item display name
category	Enum	Type of item (eg. shirt, pet)
unlockConditions	JSON	Conditions required to unlock the item

Table 10: Cosmetic Item

Attribute	Type	Description
inventoryId	UUID	Unique inventory identifier
learnerId	UUID	Learner who owns the item
itemId	UUID	The owned Cosmetic Item
source	Enum	How the item was obtained (eg. unlocked, game reward, granted by CTM)
acquiredAt	DateTime	When the item was obtained
isEquipped	Boolean	Whether the item is currently equipped

Table 11: Learner Inventory

Attribute	Type	Description
codeId	UUID	Unique access code identifier
learnerId	UUID	Learner the code applies to
createdBy	UUID	CTM who generated the code
code	String	The code entered by the recipient
codeType	Enum	Device pairing or care team invite
permissions	Enum	Permissions granted to the joining CTM; care team invites only

failedAttempts	Integer	Number of incorrect entries of this code
lockedUntil	DateTime	When further attempts are allowed again; null if not locked
status	Enum	Active, used, cancelled or expired
expiresAt	DateTime	When the code stops being valid
createdAt	DateTime	When the code was generated

Table 12: Access Code

7.3 Privacy and Data Retention

Learner, sessions and CTM data are stored on a central server in Canada. A learner's paired devices only keep what is required to play: learner game configurations, learner inventory and local session data waiting to sync.

Deleted Learner profiles and CTM accounts enter a 30-day grace period after which they are permenantly deleted including all backups. The data is no longer scheduled for deletion if the account or profile is restored within the 30-day grace period. Deleting a Learner profile entails the unlinking of any paired devices and all profile access to other CTMs. Before deleting a Learner profle, the primary CTM can download the Learner data.

The system collects minimal info about Learners because they have no login credentials nor do they own a traditional account. The system has no advertising or third-party tracking of Learners. The primary CTM (eg. a parent) controls the extent of permissions and access of other CTM's regarding the Learner's configurations and data. Any changes by a CTM to a Learner's settings is logged clearly for other CTMs to see.

When creating a Learner profile, the creator confirms they are the Learner's legal guardian or have permission from the guardian to create/manage this profile. CTMs must agree to a list of terms & conditions and privacy policies upon account registration. A CTM joining a Learner's care team requires an intentional invite from the primary CTM.

All data is encrypted in transit and at rest. Passwords are stored as securely hashed passwords rather than a readable form. Any session data being collected or queued for syncing on a paired device is kept in protected app storage until syncing, then deleted off the local storage.

The system shall comply with Canada's federal PIPEDA law and the EU's GDPR law. Additionally, the system shall meet the conditions of app store rules for children's apps regarding ads, analytics and external links.

8 Non-Functional Requirements

Usability

NFR-USE-1: 90% of CTMS experience little to no difficulty with the setup process for account creation, Learner pairing, and invitation of other CTMs.

NFR-USE-2: 90% of CTMs are easily able to navigate the system with minor difficulty in locating views such as a Learner's settings or starting gameplay for a Learner.

NFR-USE-3: 90% of CTMs are more quickly able to complete these actions in following sessions compared to the first time, demonstrating the system's memorability.

NFR-USE-4: 90% of CTMs experience little to no difficulty facilitating gameplay for a Learner.

NFR-USE-5: 80% of Learners are easily able to participate in gameplay, with minor frustration or lack of understanding reported.

The above 5 requirements will be measured through surveys with CTMs and Learners and/or observance, with the process outlined in section 4.2 of the VnV Plan.

Efficiency

NFR-EFF-1: A Learner's game data can be quickly understood and analyzed.

Accuracy

NFR-ACCU-1: A Learner's data and dashboard statistics should be completely up-to-date with any gameplay that occurred before the data page was opened.

NFR-ACCU-2: The games' content and difficulty is aligned with Ontario's Elementary curriculum for grades 3-5.

Portability

NFR-PORT-1: The system should run on Apple iPads and support iPadOS 27, as well as being compatible with iPadOS 26 and 18 (Mood Up, 2018).

NFR-PORT-2: The system should run on Android Tablets and support Android 17, as well as being compatible with Android 16 and 15 (Mood Up, 2018).

NFR-PORT-3: The system is available in web-based form and should run on the following web browsers: Google Chrome, Mozilla Firefox, and Microsoft Edge.

Accessibility

NFR-ACCESS-1: The system shall use clear and simple language throughout, especially within game instructions, to support CTMs' and Learners' understanding and ease of use.

NFR-ACCESS-2: For the scope of capstone, English will be the only language supported by the system.

NFR-ACCESS-3: The system shall provide both visual and auditory feedback and output, both of which can be adjusted for Learners' specific needs.

NFR-ACCESS-4: The system shall allow for pause, retrying, and review of game instructions at any point in gameplay.

Security

NFR-SEC-1: The system shall protect CTMs' account login information.

NFR-SEC-2: The system shall protect Learners' profile information and setting configurations.

NFR-SEC-3: The system shall protect Learners' gameplay data.

The above 3 requirements will be enforced through the specifications made in Section 7.3.

Reliability

NFR-RELY-1: In-game response times to actions are consistent, making gameplay more predictable.

Performance

NFR-PERF-1: 90% of CTMs report satisfaction with response time for loading-related actions such as logging in or accessing a Learner's game-specific dashboard statistics.

NFR-PERF-2: In-game response times are sufficiently fast that they don't interfere with game logic or confuse/upset Learners.

The above 2 requirements will be measured through surveys with CTMs and Learners and/or observance, with the process outlined in section 4.2 of the VnV Plan.

Maintainability

NFR-MAINT-1: The effort required to make any of the likely changes listed for Spectropica should be less than FRACTION (defined in 14.3) of the original development time.

9 AI and Machine Learning Requirements

Not applicable. Spectropica does not include an AI or machine learning component.

9.1 Model Inputs

N/A

9.2 Model Outputs

N/A

9.3 Evaluation Metrics

N/A

9.4 Known Limitations

N/A

10 Requirements Traceability Matrices

Stakeholder Req	Use Case	Functional Req
SR-LRN-1	4, 5	FR-PERS-3
SR-LRN-3	1	FR-PLAY-2, FR-PLAY-4, FR-PLAY-5, FR-PLAY-6, FR-PLAY-7, FR-PLAY-8, FR-PLAY-9
SR-LRN-4	1, 6	FR-TMPL-3, FR-RWD-1, FR-RWD-4
SR-CUST-3		FR-PLAY-4, FR-PLAY-5, FR-PLAY-6, FR-PLAY-7, FR-PLAY-8, FR-PLAY-9
SR-CTM-2	4, 5	FR-PERS-3
SR-CTM-3	5	FR-TMPL-3
SR-CTM-4	3	FR-DATA-1, FR-DATA-2, FR-DATA-3, FR-DATA-4, FR-DATA-5
SR-CTM-5	9, 10	FR-TEAM-1, FR-TEAM-5
SR-CTM-6	10	FR-TEAM-4, FR-TEAM-7
SR-DOM-2	5, 6	FR-TMPL-3, FR-RWD-1, FR-RWD-4
SR-DOM-3	5, 6	FR-PERS-3, FR-TMPL-4
SR-DOM-4	4, 5	FR-PERS-2, FR-TMPL-5

Table 13: Traceability Between Stakeholder Requirements, Use Cases, and Functional Requirements

Functional Req	Nonfunctional Req
FR-CTM-4, FR-DATA-1, FR-DATA-9	NFR-SEC-1, NFR-USE-1
FR-LRN-3	NFR-SEC-2
FR-DATA-1, FR-DATA-9	NFR-SEC-3, NFR-EFF-1, NFR-ACCU-1
FR-PLAY-2, FR-PLAY-4, FR-PLAY-5, FR-PLAY-6, FR-PLAY-7, FR-PLAY-8	NFR-USE-4, NFR-USE-5, NFR-ACCESS-4
FR-PERS-3, FR-TMPL-4	NFR-ACCESS-3

Table 14: Traceability Between Functional and Nonfunctional Requirements

11 Rationale

Table 15 provides justification for selected requirements and scope decisions that influence Spectropica's functionality and operating environment.

Table 15: Rationale for Selected Requirements and Scope Decisions

ID	Rationale
Section 2.1	Spectropica is designed to support educational practice rather than clinical assessment. Evaluating developmental improvement or making diagnostic conclusions would require additional research and validation beyond the scope of this project.
Section 2.2	The initial system focuses on routine sequencing, functional communication, and short-term memory to establish a manageable set of learning activities that can be developed and evaluated within the capstone timeline.
Section 2.3	Separating Learners and CTMs into distinct user groups allows Learners to focus on participating in Games while CTMs manage settings, permissions, and performance information.
Section 2.4	The system considers differences in Learners' communication methods, developmental abilities, and familiarity with technology because autistic individuals may have different learning needs and preferences.

Continued on next page

ID	Rationale
SC1	Supporting commonly available devices allows Spectropica to be used in homes and educational environments without requiring specialized equipment.
SC2	Supporting standard input methods accommodates different device configurations and interaction preferences while keeping the initial development and testing scope manageable.
SC3	Network connectivity is necessary to maintain consistent Learner information, Game configurations, and care team permissions across authorized users and devices.

For functional requirements regarding CTMs, we are allowing CTM's to delete and restore their accounts or Learners for which they are authorized to do so to comply with industry standard privacy policies for handling user data.

For functional requirements regarding Learner profiles, we've decided to not give the Learner a typical login-style account. This is because we do not want the Learner, who may be young or possess untrained memory skills, to remember their login every time. Therefore, we have created a device pairing system for the Learner to play on their authorized device(s) when setting up, not requiring any further admin steps after that.

For the decision to make a primary CTM, we wanted to avoid conflicting decisions in the Care Team to affect the Learner. We also wanted to give an assured way for legal guardians to be comfortable with allowing a specialist, educator or therapist to adjust settings without feeling powerless to revert their changes or remove them from handling the Learner profile altogether. In this manner, the legal guardian has the right to remove other CTMs who are no longer affiliated with the Learner's learning path, instead inviting a new CTM for example. The granularity of the permissions the primary CTM can invoke on each CTM should further comfort a legal guardian in their control over who can affect the Learner's experience.

Regarding the extensive personalization and template feature: autistic learners specifically are more easily prone to either understimulation or overstimulation. In that case, we should not adopt a one-size-fits-all gameplay experience as a typical educational game suite would, but rather allow the CTMs that are most knowledgeable about the Learner to have stronger capability to tailor the experience for the Learner.

THe gameplay is designed to be low-pressure with the ability to pause, abort and find help all afforded to the Learner by default. This further emphasizes the experience of the Learner and the self-applied guardrails we've employed to not cause a Learner any unintended distress. The Learner's data section regarding these games is also crucial for the CTMs to assess the Learner's current skillset and for progress tracking. The reward system is another method by which we plan to boost the Learner's engagement and positive reinforcement instead of repetitive educational gameplay they may easily be bored by.

The data requirements follow suit to what the functional requirements say about the

system. The major data entities can be seen as nouns in the functional requirements for the most part. The Learner Game Configuration is a table designed to hold the Learner's configured settings for each game by the CTM's authorization. Device and Access Code are two entities that operate together in the case of the Learner's device pairing, though the Access Code table doubles as an invite system for CTMs to join a care team.

The data system is decomposed in such a way that objects created from these entities do not destroy other objects upon deletion. For example, despite the CTM account 'owning' the Learner profile, deleting the CTM account should not also delete a Learner's profile. Even with data entities such as Learner Inventory; the aim is to not create objects that are inefficient for a database to query what it requires. It is more efficient to have a Learner Inventory entity than to have either the Learner store their cosmetic items in the same table, whether they are equipped or not etc.

There are no AI requirements because we are not employing a machine learning model in Spectropica.

12 Likely Changes

The following changes are considered likely as Spectropica is developed, tested, and evaluated with intended users. Each entry identifies the part of the system expected to evolve and its relationship to existing requirements or data entities.

LC1: **Game Content and Difficulty**

The content, challenges, and available difficulty levels within Games are likely to change based on feedback from Learners, Care Team Members (CTMs), and Domain Experts. New activities or variations may be introduced, and existing challenges may be adjusted to better accommodate individual needs.

These changes are expected to affect Game content and configuration rather than the underlying relationship between a Learner and their authorized Games. Changes to content or difficulty may also affect how previously recorded results are interpreted and compared with results from later Game Sessions.

Traceability: Game (Table 6), Learner Game Configuration (Table 8), Template (Table 9), Game Session (Table 7); FR-PERS-2, FR-PERS-3, FR-TMPL-3, FR-TMPL-5, FR-DATA-7.

LC2: **Personalization and Template Settings**

The available personalization settings and types of Template modifications are likely to expand. Additional settings may be introduced to give CTMs greater control over Game content, presentation, or difficulty.

The mechanisms for assigning configurations and Templates to Learners are expected to remain applicable even as the range of customizable options changes.

Traceability: Learner Game Configuration (Table 8), Template (Table 9); FR-PERS-3, FR-PERS-4, FR-TMPL-3, FR-TMPL-4, FR-TMPL-6.

LC3: **Performance Metrics and Dashboard Representations**

The metrics recorded for Game Sessions and the methods used to present them are likely to change as CTMs provide feedback on which information is most useful.

For example, the Dashboard may be updated to emphasize different performance measures or provide additional representations of a Learner's Game history. These changes should preserve the association between performance records and their corresponding Learner and Game Session. Changes to scoring or performance measures may affect comparisons between historical and newer results and should be considered when interpreting performance over time.

Traceability: Game Session (Table 7); FR-DATA-2, FR-DATA-3, FR-DATA-4, FR-DATA-5, FR-DATA-7.

LC4: **Rewards and Cosmetic Items**

The available Cosmetic Items, rewards, and conditions for unlocking them are likely to change as the team evaluates Learner engagement. New items may be added, existing items may be modified, and reward conditions may be adjusted.

These changes affect the available content and reward conditions rather than the relationship between Learners and the items in their inventories.

Traceability: Cosmetic Item (Table 10), Learner Inventory (Table 11); FR-RWD-1 through FR-RWD-7.

LC5: **Interface Design and Accessibility Options**

The presentation and navigation of the Game Suite and CTM Dashboard are likely to change following usability evaluations. These changes may involve instructions, navigation, feedback presentation, and available accessibility settings.

Such modifications are expected to improve the user experience without changing the responsibilities of Learners and CTMs.

Traceability: Learner Game Configuration (Table 8); FR-PLAY-4, FR-PLAY-5, FR-PERS-3, FR-DATA-5. Related usability and accessibility requirements should be included when finalized in Section 8.

LC6: **Template Organization and Sharing**

The way CTMs organize and share Templates may change as the team evaluates how these features are used. Future changes may include improvements to Template folders, importing Templates, and managing customized content.

These changes are expected to preserve the purpose of Templates as reusable variations of existing Games.

Traceability: Template (Table 9); FR-TMPL-1, FR-TMPL-2, FR-TMPL-3, FR-TMPL-9.

LC7: **Platform Compatibility and Interaction Methods**

The range of supported platforms and input methods may change as Spectropica is developed and tested. The initial intended platforms include iPadOS and Android tablets, along with web access. Additional desktop platforms, such as Windows and macOS, may be considered depending on compatibility testing and available development resources.

Changes to supported platforms may also affect available input methods, including touch, keyboard, and pointer interaction. These changes are expected to expand or refine the environments in which Spectropica operates without changing the underlying relationship between Learner profiles and paired devices.

Traceability: SC1 and SC2; applicable portability and accessibility requirements in Section 8. Related device-pairing entities and requirements: Device (Table 3), Access Code (Table 12); FR-DEV-1 through FR-DEV-5.

LC8: **Additional Games and Learning Activities**

The number and variety of Games available in Spectropica may increase beyond the initial set of learning activities. New Games may introduce additional objectives, content, and performance measures.

These additions are expected to follow the existing approach of managing Game authorization, Learner-specific configurations, and Game Session records.

Traceability: Game (Table 6), Learner Game Configuration (Table 8), Game Session (Table 7); FR-PERS-1, FR-PERS-2, FR-DATA-7.

13 Unlikely Changes

The following assumptions are considered unlikely to change during the development of Spectropica. They define stable characteristics that the system's design can rely on. Each assumption is supported by the system's intended purpose, user responsibilities, or existing requirements.

ULC1 : **Distinction Between Learners and CTMs**

Spectropica will continue to distinguish between Learners, who participate in Games, and Care Team Members (CTMs), who manage and support their learning activities. The specific functions available to each group may evolve, but their primary responsibilities are expected to remain separate.

Maintaining this distinction allows Learners to focus on participating in Games without being responsible for administrative tasks, while CTMs retain responsibility for managing activities and interpreting results.

Traceability: Learner (Table 2), CTM (Table 4).

ULC2 : **Individual Learner Profiles**

Each Learner will continue to have an identifiable profile associated with their Games, configurations, and Game Session records. This relationship is fundamental to providing individualized activities and maintaining a history of performance.

Maintaining individual profiles also allows Spectropica to associate the correct configurations and records with a Learner when they use different paired devices or receive support from multiple CTMs.

Traceability: Learner (Table 2), Learner Game Configuration (Table 8), Game Session (Table 7); FR-LRN-1, FR-DATA-7.

ULC3 : **Care Team Membership and Permissions**

Spectropica will continue to associate CTMs with Learners through care team memberships. Each active Learner profile will have a Primary CTM responsible for managing the permissions of other CTMs. This structure establishes clear responsibility for access management when multiple individuals support the same Learner.

Although the available permissions may change, the distinction between authorized and unauthorized actions is expected to remain. The identity of the Primary CTM may change through the ownership-transfer process without changing the purpose of the role.

Traceability: CTM (Table 4), Care Team Membership (Table 5); FR-LRN-2, FR-TEAM-4, FR-TEAM-6, FR-TEAM-7.

ULC4 : **CTM-Controlled Personalization**

CTMs will continue to manage Learner-specific Game settings and authorizations. The system is expected to preserve the principle that personalized activities are configured by authorized CTMs rather than automatically prescribed based on assumptions about a Learner's developmental needs.

This approach maintains CTM responsibility for selecting activities appropriate to individual Learners, even if the range of available personalization settings expands.

Traceability: Learner Game Configuration (Table 8), Template (Table 9); FR-PERS-1 through FR-PERS-4.

ULC5 : **Games as the Primary Learning Experience**

Games will remain the central activities through which Learners interact with Spectropica. The content and number of Games may change, but their role as the primary means of providing practice opportunities is expected to remain.

Replacing Games with a different primary learning format would fundamentally change Spectropica's intended purpose rather than simply extend its functionality.

Traceability: Game (Table 6), Game Session (Table 7).

ULC6 : **Game-Specific Performance Records**

Spectropica will continue to record performance information associated with individual Learners and Game Sessions. The metrics and methods of presentation may evolve, but the relationship between a Learner, a Game, and its recorded session results is expected to remain.

Maintaining these relationships allows CTMs to review individual Game histories and interpret recorded results in the context of the activities completed.

Traceability: Learner (Table 2), Game (Table 6), Game Session (Table 7); FR-DATA-2, FR-DATA-4, FR-DATA-7.

ULC7 : **Learner Control Over Participation**

Learners will continue to have the ability to pause, restart, or end an ongoing Game. These controls allow Learners to stop or repeat an activity according to their needs and preferences, making them an important part of the intended gameplay experience.

Removing these controls would reduce the flexibility available to Learners during participation and would conflict with the existing gameplay requirements.

Traceability: FR-PLAY-5, FR-PLAY-6, FR-PLAY-7.

ULC8 : **Non-Clinical Purpose**

Spectropica will remain an educational support application rather than a clinical assessment or diagnostic tool. Performance information will continue to describe outcomes within Games rather than establish broader developmental conclusions.

Changing this purpose would require a substantial reconsideration of the system's scope and validation requirements.

Traceability: Game (Table 6), Game Session (Table 7).

14 Appendices

14.1 Appendix A: Additional Materials

Figure 2 shows the state diagram for playing a Game, including loading, gameplay, pausing, rewards, restarting, and exiting.

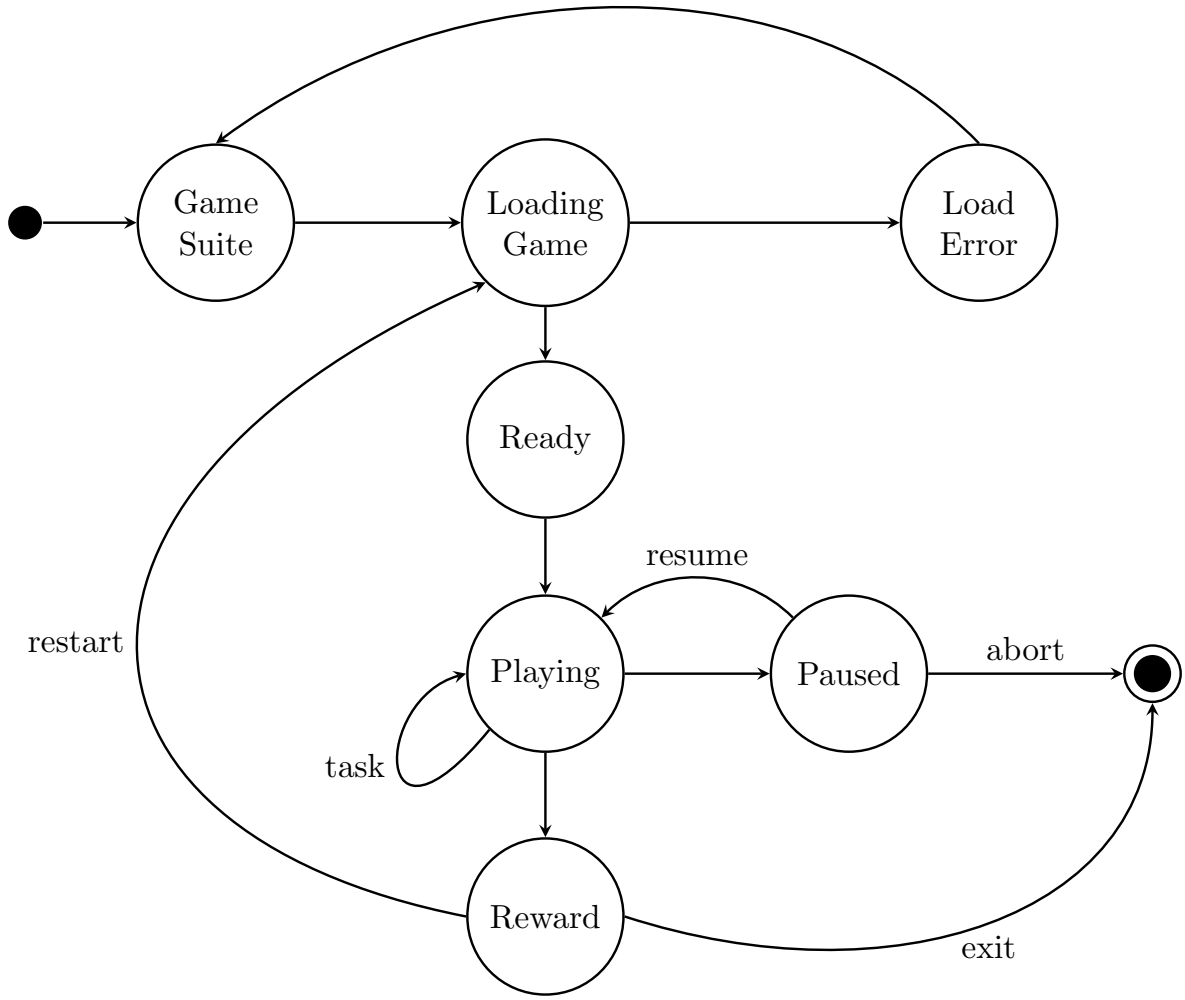


Figure 2: State diagram for playing a Game.

14.2 Appendix B: Development Plan

The project's development process and implementation plan are documented in the Development Plan in the project repository.

14.3 Appendix C: Values of Auxiliary Constants

The following auxiliary constant is used in the nonfunctional requirements. Table 16 lists its value and referenced requirement.

Constant	Value	Referenced Requirement
`FRACTION`	1/6	NFR-MAINT-1

Table 16: Values of Auxiliary Constants

14.4 Appendix D: Generative AI Disclosure

Artificial Intelligence (AI) Tools Used: Claude's Sonnet 5 model and ChatGPT (OpenAI).

14.4.1 How and Where Used

ChatGPT (OpenAI) assisted with a final grammar and spelling review of portions of this Software Requirements Specification (SRS).

Sonnet was used to learn how to resolve LaTeX errors and structure LaTeX documents.

We used AI to convert our team-created system context diagram into a vector diagram using LaTeX/TikZ. We also used AI to add citations to the text and format the corresponding BibTeX entries.

14.4.2 Verification of AI-Generated Content

AI usage was limited to resolving LaTeX errors, formatting citations, structuring the document, converting the team-created system context diagram into a vector diagram, and reviewing grammar and spelling.

14.5 Appendix E: Reflection

1. **What went well during the development of the SRS?**

   One aspect that went well was our team's ability to build on our initial project proposal and develop a more detailed understanding of what Spectropica needs to accomplish. As we worked through the Software Requirements Specification (SRS), we were able to break down our overall project goals into more specific system functionalities and identify how different parts of the application would interact. Having a shared understanding of our project's purpose helped us organize the requirements and maintain consistency between sections. We also benefited from being able to discuss different ideas as a team and refine our decisions as the project became more clearly defined.

2. **What challenges did you experience, and how did you resolve them?**

   One of the main challenges we experienced was distinguishing between high-level project goals and specific system requirements. Initially, it was difficult to determine how much detail belonged in each section and how to describe what the system needed to accomplish without prescribing exactly how it would be implemented.

   Another challenge was ensuring that the requirements remained consistent across different sections of the SRS, especially as our understanding of the project evolved. We addressed these challenges by referring back to the SRS template, reviewing the requirements as a team, and comparing them against our overall project objectives. This helped us make our requirements more specific while ensuring they continued to support the intended purpose of Spectropica.

3. **How many requirements were inspired by discussions with clients, stakeholders, or potential users?**

   Approximately ten of our requirements were influenced by discussions with our project supervisor, who is our domain expert in psychology and autism. These discussions helped us better understand the different needs and abilities of autistic Learners, particularly the importance of personalization, accessibility, and flexibility during gameplay. This influenced requirements related to Game customization, adjustable settings, clear instructions, and allowing Learners to control their participation. The discussions helped us ensure that our requirements considered the needs of our intended users rather than focusing solely on technical functionality.

4. **Which courses helped prepare your team for this capstone deliverable?**

   SFWRENG 3RA3, our Software Requirements course, was especially useful for developing this deliverable. The course introduced us to the process of identifying, organizing, and documenting software requirements, which directly applied to writing our SRS. In particular, it helped us understand the distinction between functional and nonfunctional requirements, the importance of clear and unambiguous wording, and how to structure requirements so that they can be understood and evaluated.

   The knowledge gained from this course provided a strong foundation for translating our project idea into a more formal specification.

5. **What additional knowledge or skills do you need to acquire?**

   Amirah Abdel-Rahman:

   One area where I would like to develop a stronger understanding is developmental psychology, particularly autism and how differences in cognitive, communication, and learning abilities may affect the way individuals interact with educational games. Since Spectropica is designed for autistic Learners, I think it is important to understand these differences so that the decisions we make are appropriate for our intended users rather than based on assumptions.

I would also like to improve my knowledge of game design, especially how to create engaging and accessible activities that balance learning objectives with an enjoyable user experience. While I have experience with software development, designing educational games for a specific user group introduces considerations that I have not explored in as much depth.

Marko: I have some knowledge in game development, but I currently lack the hands-on experience with database management specifically for a game engine context. I understand SQL and security well enough, as well as game engines like Godot, but as for bridging the gap between them, I am unfamiliar with the best practices.

Maya Azar:

One area where I need to develop a stronger understanding is autism, particularly the different needs, abilities, and challenges that autistic individuals may experience. Currently, my knowledge is mostly surface-level, so I think it is important to learn more about autism to ensure that the Games we develop are appropriate for our intended users. I would also like to gain a better understanding of how children learn and develop new skills, especially how different learning approaches can influence their engagement and progress. Since Spectropica is designed to support learning, having a stronger foundation in these areas will help me make more informed design decisions.

Meena Al Omran:

I have a lot to learn technically. I don't have any experience in game design, although I've always found it interesting. It will be new to program real-time interactions and to also consider things like enjoyability, rewards, etc. Most of my experience is very objective-based. I also have never designed for an app before and have only worked in web development, so that will be a new environment to work in. I also have a ways to go with understanding the needs and experiences of autistic children, although I do have some basic knowledge from research I've done for personal interest. Most of my knowledge of autism is about how adults experience it. It will be new to consider a child's point of view and how to help them. I look forward to the challenge.

Omar Ghoneim:

Despite having a C++ background, I am not familiar with game engines and their quirks. I am also not experienced in the field of game physics and how to draw good art or visuals in the context of gaming. This will also be my first time dealing with very sensitive data and integrating heavy privacy and security mechanisms.

I also need to acquire the macrosystem understanding of backend engineering, with the game engine, with the frontend and the account management altogether wrapped up. I need to acquire the skill of planning a system that expansive and complex.

6. **How will you acquire these skills, and which approach will you choose?**

   Amirah Abdel-Rahman:

To strengthen my understanding of autism and developmental abilities, I plan to continue reviewing academic research on autism and educational interventions and consulting with our domain expert to gain more practical insight into the needs of our intended users. Consulting our expert will allow me to ask questions directly related to our project and receive feedback on our design decisions. Reviewing research will help me build the foundational knowledge needed to make those discussions more meaningful.

For game design, I plan to prioritize hands-on prototyping because I learn best by applying concepts directly. This will allow me to explore different game mechanics, identify usability issues, and gradually improve my understanding of how to design activities that are both engaging and accessible.

Marko: A good approach is online research and finding learning resources for this problem specifically. I could also personally ask actual industry professionals with past experience in the area how they would do it. And then, I could practice on a mini side-project that uses Godot and bridges data to an online SQL hosting service, so that I get hands-on mastery of the skill.

Maya Azar:

To improve my understanding of autism and children's learning processes, I could review academic research and educational resources or speak with individuals who have professional or personal experience in these areas. I plan to use a combination of both approaches. Research will help me build a stronger foundation and understand different learning needs, while speaking with experts, such as our project supervisor, will allow me to ask questions specific to Spectropica. I also plan to learn from friends and family members with relevant experience, including those who work as special education teachers. I think these conversations will be especially valuable because they can provide practical insight into how children with different learning needs respond to educational activities, which can help guide our Game design decisions.

Meena Al Omran:

To improve my technical knowledge, I will have to do some research about game development and how it differs from the type of programming I'm used to. It will also help to talk to my groupmates who have more experience in game development about how to learn and what goes into it. It would help to do some sample coding on a much smaller scale before we start implementation. The same goes for app development - I will need to do some reading and try out some small examples. I plan to improve my knowledge of autism, specifically autistic children, by reading research. Our supervisor will also be a big help in understanding not only their development, but general cognitive development in children. I think that also speaking to educators in the field will help me gain some perspective and understand their needs better.

Omar Ghoneim:

I will acquire the skill of dealing with the Godot engine by doing a volume of demonstrations and practice projects on a small scale with Godot, each harnessing different areas important to Spectropica such as game physics, backend engineering and frontend design. I will try to agglomerate these into one skillset that can be used for a large full-stack project like Spectropica.

Regarding the macrosystem understanding and planning, I will ask some colleagues and friends I have that are veterans in the field of designing large systems and handling sensitive data, of which some are older and more experienced. I will create mock plans for mock project to practice the skill of system decomposition and system design, to then apply it the Spectropica context.

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